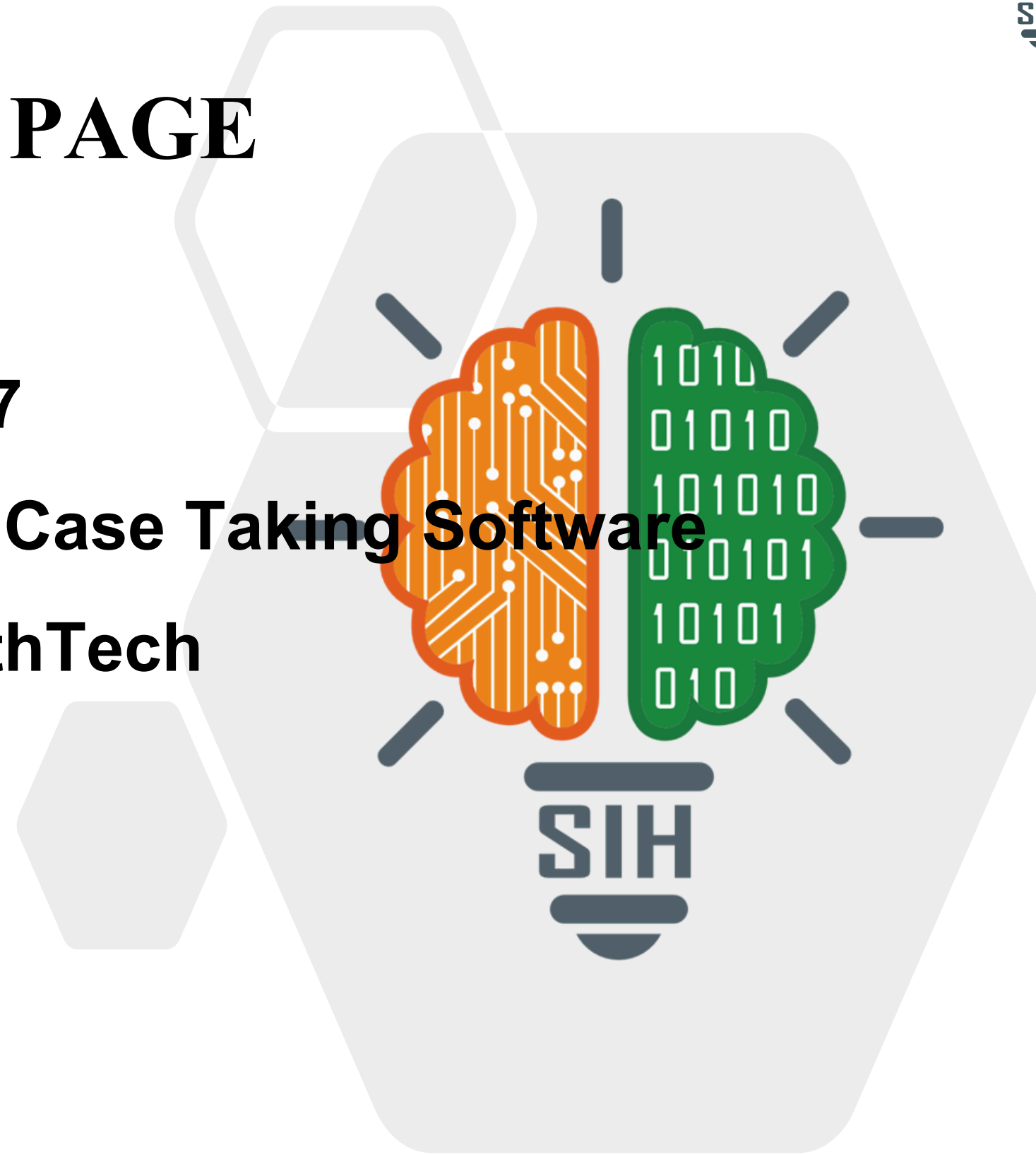


TITLE PAGE

- Problem Statement ID – SIH26047
- Problem Statement Title- Patient Case Taking Software
- Theme- MedTech /BioTech /HealthTech
- PS Category- Software
- Team ID-
- Team Name – Team Random



Problem

- Fragmented & Unstructured Medical Records.
- Registration-Focused, Not Clinical History Focused.
- Low Accessibility for Elderly, Rural & Low-Literacy Patients.
- Manual History-Taking Doesn't Scale for High OPD Loads.

Our Solution

- Adaptive AI Clinical Interview
- History + Medications in One View
- Physician-Ready Patient Summary
- Reports & Prescriptions, Unified
- Red-Flag Triage
- HIS / EMR / ABHA Integration
- Dedicated Patient, Lab & Physician Views

Why Different

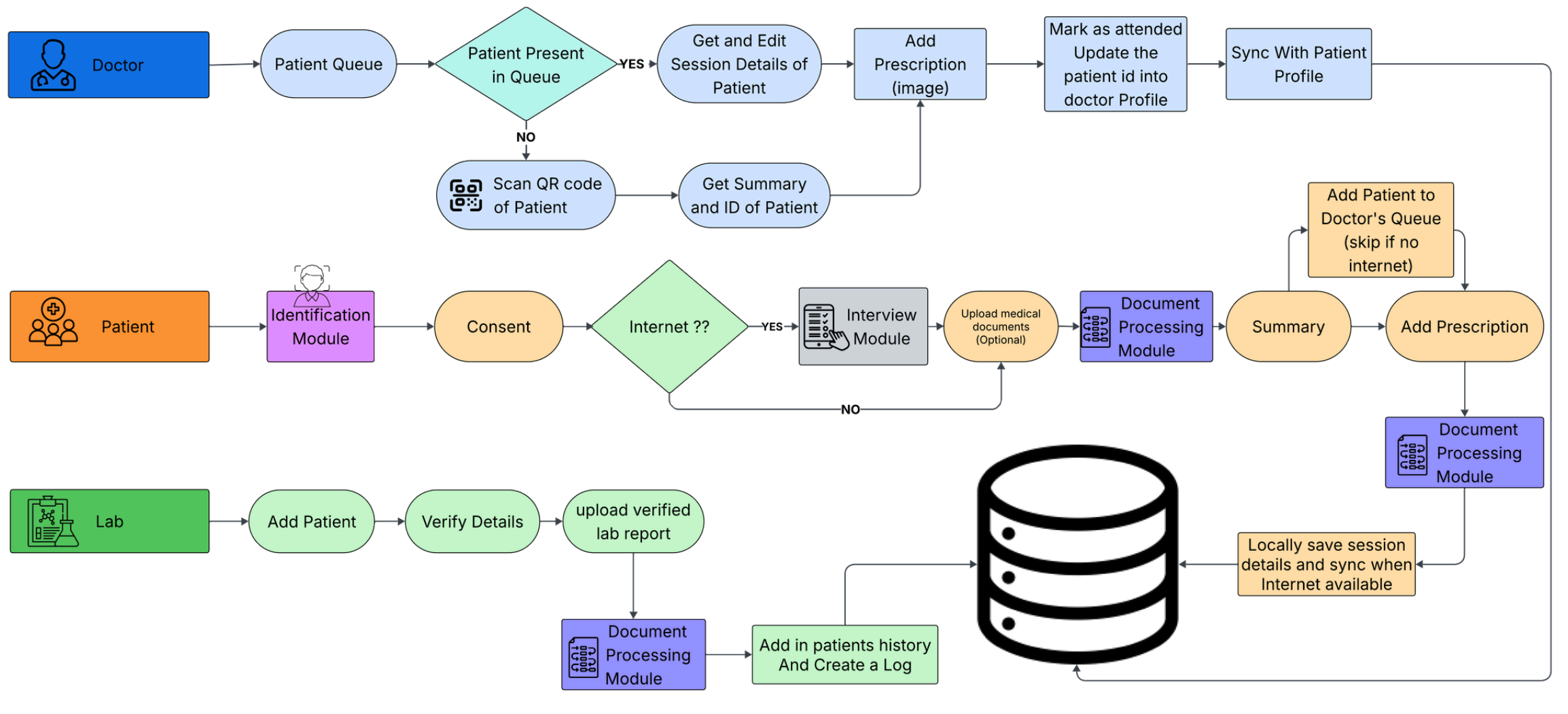
- Voice + Touch + Multilingual
- Offline-Ready Access
- Emergency Detection → Automated SOS Call
- Voice based Mental Health Detection
- Consent-First, Privacy-by-Design
- Smart Report & Doctor Prescription Extraction + Specialist Verification

Key Value Propositions

- Summarized history ready before consultation
- Unified timeline of previous medical records
- Accessible Across Languages & Literacy Levels
- Scalable for high-volume government OPDs

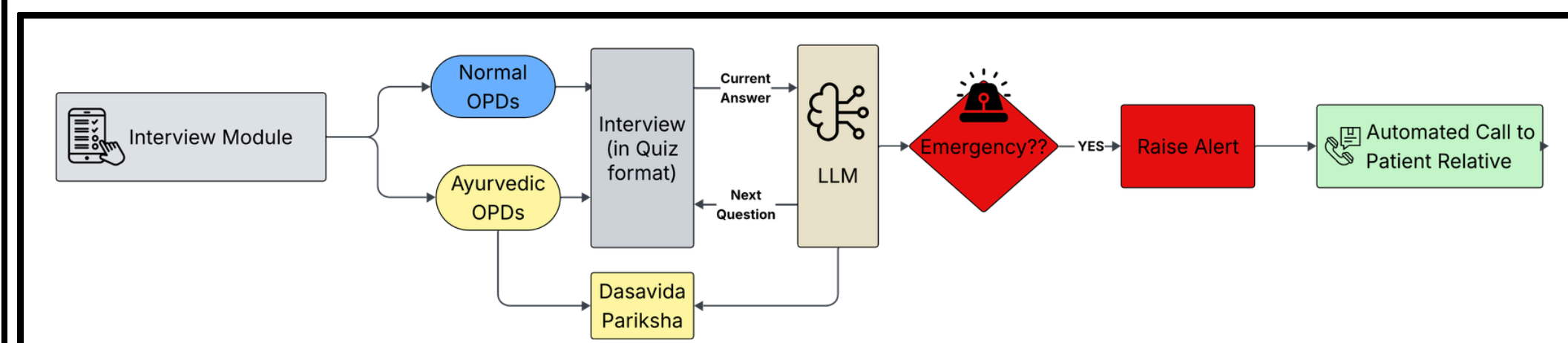
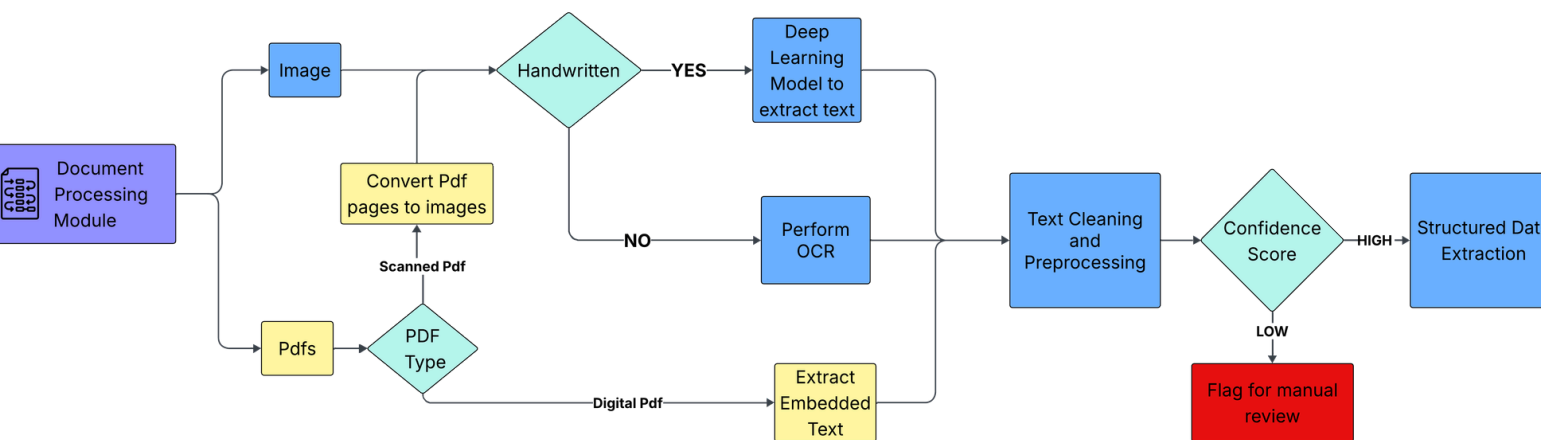
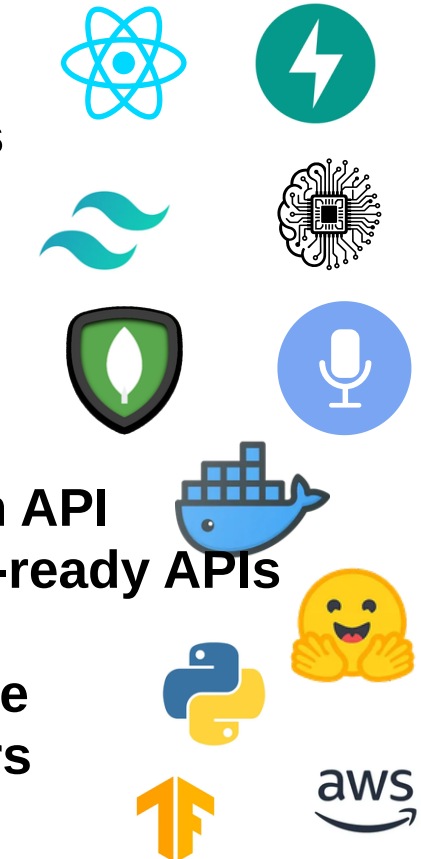
Patient Journey Flow





TECH STACK

- React
- FastAPI
- Tailwind css
- JWT
- Python
- MongoDB
- Groq API
- OCR
- Web Speech API
- ABDM/FHIR-ready APIs
- Docker
- Hugging face
- Transformers
- AWS



Risk Assessment and mitigation

LLM dependency for critical decisions

Hybrid red-flag detection (rule-based + LLM) for safety-critical triage

No internet/ API access

Saved Summary and Scripted fallback interview in local storage

Handwrittend Prescription recognition

Deep Learning Model + Specialist approval

Interview may be overhead for patient in pain

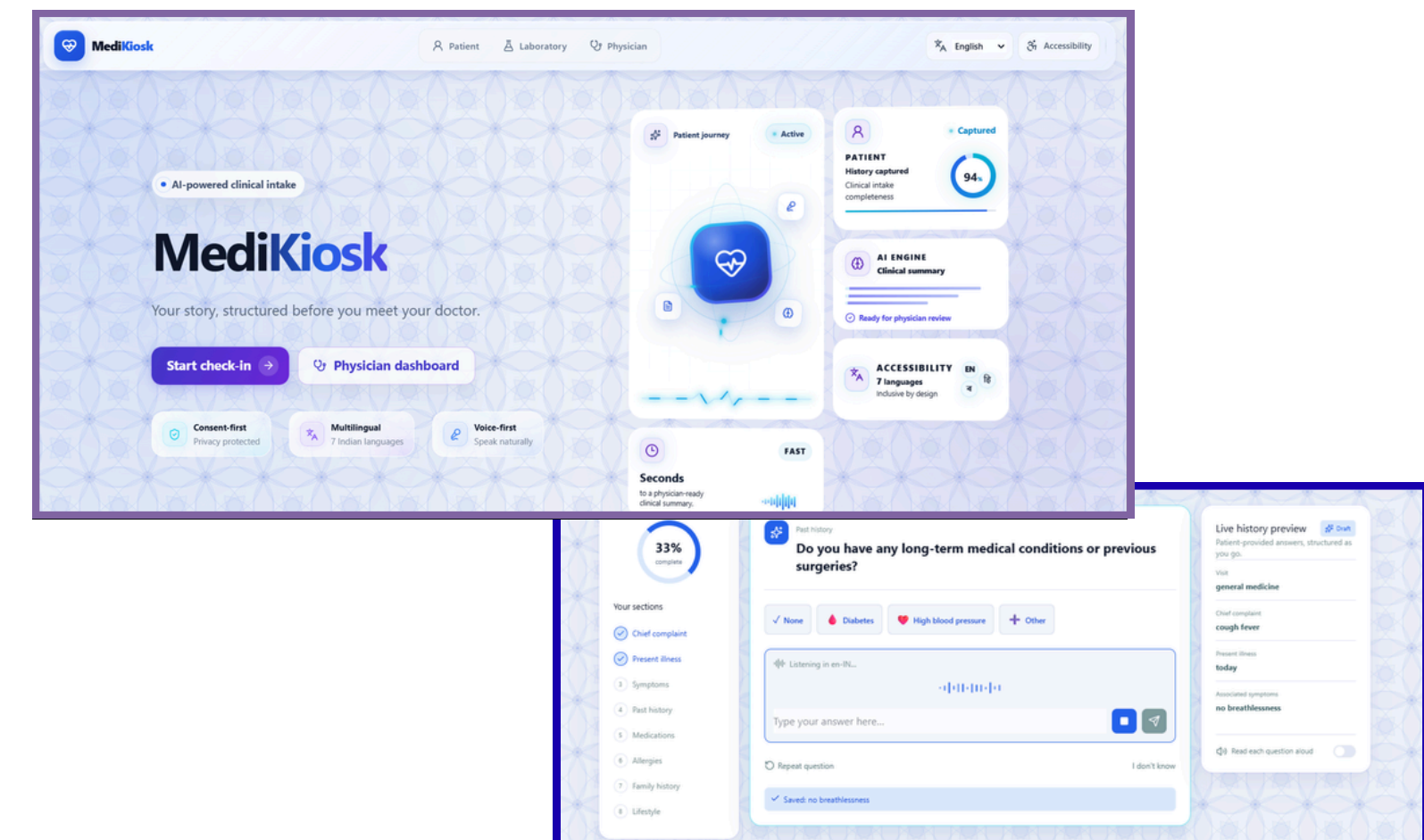
Patient can book direct consultation with doctor; red flag triage in emergency

Data privacy & duplicate patient records

unique indexes on Aadhaar/mobile, consent-gated data flow, no full Aadhaar returned to clients

Technical Feasability

- React 19 + TypeScript , FastAPI + MongoDB backend, AWS for document storage — modern, well-supported stack
- Adaptive interview engine driven by editable clinical_ontology.json (no hard-coded logic)
- Voice input/output via Web Speech API for low-literacy/accessibility use and available in multiple languages.
- OCR pipeline for prior report digitization; Specialized deep learning models for doctor handwritten prescription recognition



Resource Requirement

- Standard cloud/on-prem servers (no GPU-heavy inference needed for MVP)
- Open-source stack: MongoDB, FastAPI, React — no proprietary licensing cost
- Advanced LLMs for better interviews & summaries → higher API/inference costs
- Kiosk-grade touchscreen hardware for patient-facing deployment

Patient Impact

- Reduces OPD waiting time via pre-elicitation and history management.
- Multimodal (voice + touch) interview improves accessibility for elderly/low-literacy patients
- Multilingual support widens reach across diverse patient populations
- Faster identification of urgent cases via real-time red-flag alerts

Clinician Benefits

- AI-assisted clinical summary cuts documentation time per patient
- Structured queue with pre-collected history improves consultation quality
- Deterministic red-flag detection ensures safety-critical alerts aren't missed
- Physician dashboard centralizes patient records, prescriptions, and history

Healthcare System Impact

- Reduces patient load bottleneck at OPD front desks
- Improves continuity of care with persistent per-visit clinical records
- Digitizes prior reports/prescriptions, reducing paper dependency
- Scalable across different cities and facilities via configurable locations parameter

Social and Economic Benefits

- Lowers effective cost-per-consultation by automating intake
- Enables better resource allocation in high-patient-volume public hospitals
- Supports India's digital health mission (ABDM-aligned integration)
- Subscription/SaaS model creates sustainable revenue for scale-up

Research Papers

- Handwritten medical prescription recognition
- MedASR
- Personalized Healthcare Management
- Automated phone calling

Market Research

- <https://www.grandviewresearch.com/industry-analysis/medical-document-management-systems-market>
- <https://pmc.ncbi.nlm.nih.gov/articles/PMC10440204/>
- <https://cis-india.org/internet-governance/blog/health-data-management-policies>
- <http://pmc.ncbi.nlm.nih.gov/articles/PMC6033126/>

- PROTOTYPE LINK : <https://medikiosksih26.vercel.app/>
- GITHUB LINK: <https://github.com/Vivek1-coder/sih2026>
- VIDEO LINK : <https://youtu.be/qd4GCD7sh18>